

# Formula Notes — Errata

Corrections to the 13-page handwritten *M2 unit 1-5 formulas*. **Verdict: the notes are good and on the right syllabus** — they cover all 28 sessions of MAC21 and contain nothing off-syllabus. But there are **10 real errors**, four of which would break a whole 7-marker. Fix those four in pen before you use the notes again.

**What is right:** Unit 2 entirely (circuits, cooling, orthogonal trajectories, Taylor/Euler/RK4) · the whole of numerical differentiation on p9 (I checked every coefficient symbolically — all correct) · trapezoidal and both Simpson rules · Gauss-Seidel · all four eigenvalue property proofs on p13 · variation of parameters · CF cases and P.I. types 1–3.

## Fix these four first — each breaks a full question

| PG | TOPIC                                                                   | NOTES SAY                                                                       | SHOULD BE                                                                                                                                                                                                      |
|----|-------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| p6 | P.I. Type 4 — the shift rule<br>breaks every $e^{(ax)} \cdot V$ problem | $\frac{1}{f(D)} \cdot e^{(ax)} \cdot V = e^{(ax)} \cdot \frac{1}{f(D)} \cdot V$ | $= e^{(ax)} \cdot \frac{1}{f(D+a)} \cdot V$<br>Every $D$ becomes $D+a$ . Check: for $(D-1)y = xe^{2x}$ the true P.I. is $(x-1)e^{2x}$ . The notes' rule gives $(-x-1)e^{2x}$ — wrong.                          |
| p7 | Legendre substitution<br>appeared 4x                                    | $t = \log(ax+b)$ , then $ax + b = \log t$                                       | $t = \log(ax+b)$ , then $ax + b = e^t$<br>Self-contradictory as written. Cauchy on the same page has it right ( $x = e^t$ ).                                                                                   |
| p8 | Divided difference<br>appeared 5x/5                                     | $[x_0, x_1, x_2] = ([x_1, x_2] - [x_0, x_1]) / (x_2 - x_0)$                     | $= ([x_1, x_2] - [x_0, x_1]) / (x_2 - x_0)$<br>Denominator spans <b>first to last</b> node. On the recurring PYQ data (2,4)(4,56)(9,711) the correct value is <b>15</b> ; the notes' version gives <b>21</b> . |
| p2 | N-R, simultaneous equations<br>appeared 4x                              | $-g(x_0, y_0) = h \cdot g_x(x_0, y_0) + k \cdot f_y(x_0, y_0)$                  | $-g(x_0, y_0) = h \cdot g_x(x_0, y_0) + k \cdot g_y(x_0, y_0)$<br>Second equation must use <b>g</b> throughout, not <i>f</i> .                                                                                 |

## Also wrong — fix if you have five minutes

| PG | TOPIC                                 | NOTES SAY                                                   | SHOULD BE                                                                                            |
|----|---------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| p5 | Trig identity <b>used in P.I.</b>     | $2 \sin x \cdot \sin y = \cos(x+y) - \cos(x-y)$             | $= \cos(x-y) - \cos(x+y)$ signs swapped                                                              |
| p2 | Taylor, 2 variables — 3rd degree term | $3(x-a)^2(x-b) \cdot f_{xxy} + 3(x-a)(x-b)^2 \cdot f_{xyy}$ | $3(x-a)^2(y-b) \cdot f_{xxy} + 3(x-a)(y-b)^2 \cdot f_{xyy}$<br>( $x-b$ ) written twice for ( $y-b$ ) |
| p2 | Maclaurin, 2 variables — last term    | $y^3 \cdot f_{xxx}(0,0)$                                    | $y^3 \cdot f_{yyy}(0,0)$                                                                             |
| p2 | r-s-t test — first line only          | $rt - s^2 > 0$ and $r > 0$ — minimum                        | $rt - s^2 > 0$ the other three lines are correct                                                     |
| p4 | Euler's method                        | $y_{(n+1)} = y_0 + h \cdot f(x_n, y_n)$                     | $y_{(n+1)} = y_n + h \cdot f(x_n, y_n)$ $y_0$ only works for the first step                          |
| p8 | Newton backward — general term        | $p(p+1)(p+2) \dots (p + (n-1))$                             | $p(p+1)(p+2) \dots (p + (n-1))$ early terms are fine                                                 |

## Sloppy but harmless — just know what is meant

| PG  | WHAT IT SAYS                                             | READING                                                                                                                                                                                                                                       |
|-----|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| p12 | $ A - \lambda I  X = 0$                                  | Two different things merged. Characteristic equation is $ A - \lambda I  = 0$ (a determinant); eigenvectors come from $(A - \lambda I)X = 0$ (a matrix equation).                                                                             |
| p12 | Rayleigh "determines largest <i>eigen vector</i> "       | Largest <b>eigenvalue</b> — and its corresponding eigenvector.                                                                                                                                                                                |
| p5  | $\sin^2 x = (1 - \cos^2 x)/2$                            | Make sure that reads <b>cos 2x</b> , not $\cos^2 x$ — the handwriting is ambiguous. Same for $\cos^2 x$ .                                                                                                                                     |
| p6  | Variation of parameters headed with a 3rd-order equation | The Wronskian formula given is the 2nd-order one — which is what the exam asks. Ignore the header.                                                                                                                                            |
| p12 | Matrix-method sketch with $e^t$ and $e^{-t}$ mixed       | Exponents must be <b><math>e^{(\lambda_1 t)}</math></b> and <b><math>e^{(\lambda_2 t)}</math></b> consistently. The boxed $X = P \cdot \text{diag}(e^{(\lambda_1 t)}, e^{(\lambda_2 t)}) \cdot [c_1 \ c_2]^T$ above it is correct — use that. |
| p13 | Property 4: "eigenvalues of $mA$ "                       | of <b>A</b> . The proof itself is correct.                                                                                                                                                                                                    |

## Gaps — on your syllabus, missing from the notes

**1. Finite-difference operator relations** (session 18) — nowhere in the notes. You need  $\Delta^2 y_n = y_{(n+2)} - 2y_{(n+1)} + y_n$  and  $\nabla^2 y_n = y_n - 2y_{(n-1)} + y_{(n-2)}$ . The 2-marker "express  $\Delta^2 y_n$  in terms of  $y$  values" has appeared **twice**, at slot Q7(a). Also worth knowing:  $\Delta = E - 1$ ,  $\nabla = 1 - E^{-1}$ .

**2. Legendre operator relations** — p7 gives the Cauchy ones ( $xy' = Dy$ ,  $x^2 y'' = D(D-1)y$ ) then leaves the Legendre box blank. Write them in:  $(ax+b)y' = aDy$ ,  $(ax+b)^2 y'' = a^2 D(D-1)y$ ,  $(ax+b)^3 y''' = a^3 D(D-1)(D-2)y$ . *Forgetting the  $a^2$  is the classic way to lose this question.*

**3. The inconsistent case** — p11 lists unique and infinite solutions but not  $p(A) \neq p(A:B) \rightarrow$  **no solution**. That is one of the three cases the 2-marker asks for.

**4. Two techniques with no formula to memorise**, so they are absent by nature but still examinable: **missing terms** (set the higher differences to zero) and **grouped-frequency estimation** (build the cumulative "less than" column first). Both are in the solutions PDF, §23 and §24.

## Overall

**Use them.** Coverage is genuinely complete against the MAC21 portion — all five units, and nothing from the 25MAC21 scheme (no multiple integrals, no Laplace) has crept in, so they are matched to your paper. The proofs on p13 are correct and one of them ( $\lambda \rightarrow 1/\lambda$  for  $A^{-1}$ ) was asked in July 2024.

**But correct the four red items in pen before you revise from them again** — the shift rule and the divided-difference denominator in particular are used in questions that have appeared in nearly every paper.

Checked page by page against the MAC21 session-wise portion. Every disputed formula was verified symbolically or numerically before being listed as an error — the shift rule, divided-difference denominator and trig identity were each tested on a worked example. Page 9 (numerical differentiation) was verified coefficient by coefficient and is entirely correct.